==========

STANDARD OPERATING PROCEDURE: PRINTING PRODUCTION

==========

DOCUMENT ID: SOP-PRNT-004

DEPARTMENT: Fabric Processing & Printing Division

1. PURPOSE

To standardize the pre-production, printing, and inspection workflow to maintain zero-defect output across digital, rotary, and screen print lines.

2. PROCEDURE STEPS

Step 2.1: Design Ingestion and Mapping

* The design studio delivers digital layouts to the machine operator via the local network.

* The operator checks image dimensions against target fabric width (e.g., 44 inches or 56 inches).

* Color channels are calibrated using spectrophotometers to match specified Pantone references.

Step 2.2: Fabric Pre-Treatment

* Before printing, natural fabric bases (Cotton, Silk, Viscose) must pass through a padding machine applying specialized chemical coating fixatives.

* This ensures maximum ink absorption, vibrant color retention, and washing fastness.

Step 2.3: Machine Calibration & Trial Strike-Off

* The operator runs a 2-meter test swatch (Strike-Off) on the designated machine line.

* The Shift Manager checks this sample against the master physical swatch under standardized

D65 lighting cabins.

* Production cannot proceed until the Shift Manager signs off on color matching.

Step 2.4: Mass Production Run

* Upon approval, bulk printing begins.

* Operators must monitor ink levels and print-head performance every 500 meters to

prevent banding or skipped patterns.

Step 2.5: Post-Print Curing and Steaming

* Printed fabrics must immediately pass through loop steamers at specified temperatures

(e.g., 102°C–170°C depending on fabric chemistry) to permanently lock in the dyes.

* Fabrics are washed to eliminate residual processing chemicals, then dried via stenter machines.

Step 2.6: Final 4-Point Inspection & Packaging

* Finished fabric is passed over illuminated inspection tables.

* Defect points are tallied using the International 4-Point Quality System.

* Any roll exceeding 20 penalty points per 100 square meters is downgraded to "B-Grade".

* Approved rolls are wrapped in heavy poly-sheets, barcoded, and cataloged for dispatch.